



Sistemas e Arquiteturas de Computadores

2026

André Pereira

Background to this course
(slides from several sources)



- **Documentation & oral lectures/classes**
 - *Documentation (slides, lab guides, books, reports...): English*
 - *Oral presentations: optional, but usually Portuguese*
- **Expected background**
 - *Imperative Programming*
 - *Lab skills in Unix, C/C++ programming and debugging*
- **Assessment**
 - *2x Written test*
- **Communicate with me only through `ampereira@di.uminho.pt`**

Focus on this course: cover the bases of performance engineering



How:

- understanding the organisation of a computer system (its architecture)
- plan and code efficient algorithms & data structures
- profiling & evaluating (through measurements) the execution efficiency

Where in the hardware:

- in sequential code and memory hierarchy
 - Revisiting some background concepts
- w/ support for thread parallelism
 - new threading technologies (C++ threads – let 's avoid pthreads)
 - multithreading in multiple devices
- w/ computing accelerators (GPU, ...)
 - using them simultaneously with CPUs
- w/ support for task parallelism
 - multiprocessing in a computer cluster



Tentative Course Plan



semanar	Data	Teórica		Prática Laboratorial	
1	15/09 a 20/09				
2	22/09 a 27/09	Apresentação. Introdução à arquitetura von Newman. Execução de instruções num computador.	2h	Apresentação do formato das PLs.	1h
3	29/09 a 04/10	Compilação - de HLL a LLL.	1h15	Lab 1 - Níveis de abstração	1h45
4	06/10 a 11/10	Medição de desempenho. Hierarquia de memória.	1h15	Lab 2 - Métricas de desempenho	1h45
5	13/10 a 18/10	Hierarquia de memória - continuação.	1h15	Lab 3 - Hierarquia de memória	1h45
6	20/10 a 25/10	Super escalaridade.	1h15	Lab 4 - Pipelining	1h45
7	27/10 a 01/11	Processamento vetorial.	1h15	Lab 5 - Super escalaridade	1h45
8	03/11 a 08/11	Teste 1	3h	-	0h
9	10/11 a 15/11	Arquitetura multicore.	1h15	Lab 6 - Vetorização	1h45
10	17/11 a 22/11	Paralelização com OpenMP.	1h15	Lab 7 - OpenMP	1h45
11	24/11 a 29/11	Temas relevantes para ACA (subconjunto dos conteúdos programáticos)	0h	Temas relevantes para ACA (subconjunto dos conteúdos programáticos)	0h
12	01/12 a 06/12		0h		0h
13	08/12 a 13/12		0h		0h
14	15/12 a 20/12		0h		0h
		Teste 2	3h	-	0h



- **Theoretical lessons**
 - Typically, 1-2h in length, depending on the subject
 - Hopefully interactive (depends on you!)
- **Break**
- **Practical lessons**
 - Hands-on exercises on the theoretical topics
 - Laptop to compile/run code + access to cluster (preferably SeARCH, not Deucalion)
 - Not required for only a couple of classes
 - **No support for Windows will be given in any of my classes**
 - Duration depends on the time theoretical classes take
- **Participation is not mandatory but strongly encouraged**

An Issue with the Course Schedule



	segunda-feira	terça-feira	quarta-feira	quinta-feira	sexta-feira
09:00	Computação Paralela [CG - Edifício 1 - 0.08] T1	Sistemas e Arquiteturas de Computadores [CG - Edifício 7 - 1.09] T1		Ferramentas de Computação de Elevado Desempenho [CG - Edifício 7 - 1.10] T1	Arquiteturas Avançadas de Computadores [CG - Edifício 7 - 1.09] PL1
10:00		Sistemas e Arquiteturas de Computadores [CG - Edifício 7 - 1.09] PL1		Ferramentas de Computação de Elevado Desempenho [CG - Edifício 7 - 1.10] PL1	Arquiteturas Avançadas de Computadores [CG - Edifício 7 - 1.09] PL1
11:00					
12:00					
13:00					Visualização Científica [CG - Edifício 1 - 1.43] T1
14:00	Computação Paralela [CG - Edifício 2 - 2.05] PL6	Técnicas de Classificação e Agrupamento de Dados [CG - Edifício 16 - 1.06] T1	Simulação Numérica em Engenharia [CG - Edifício 16 - 1.06] T1		Visualização Científica [CG - Edifício 1 - 1.43] PL1
15:00		Técnicas de Classificação e Agrupamento de Dados [CG - Edifício 16 - 1.06] PL1			
16:00			Simulação Numérica em Engenharia [CG - Edifício 16 - 1.06] PL1		
17:00					

Key textbook (1)

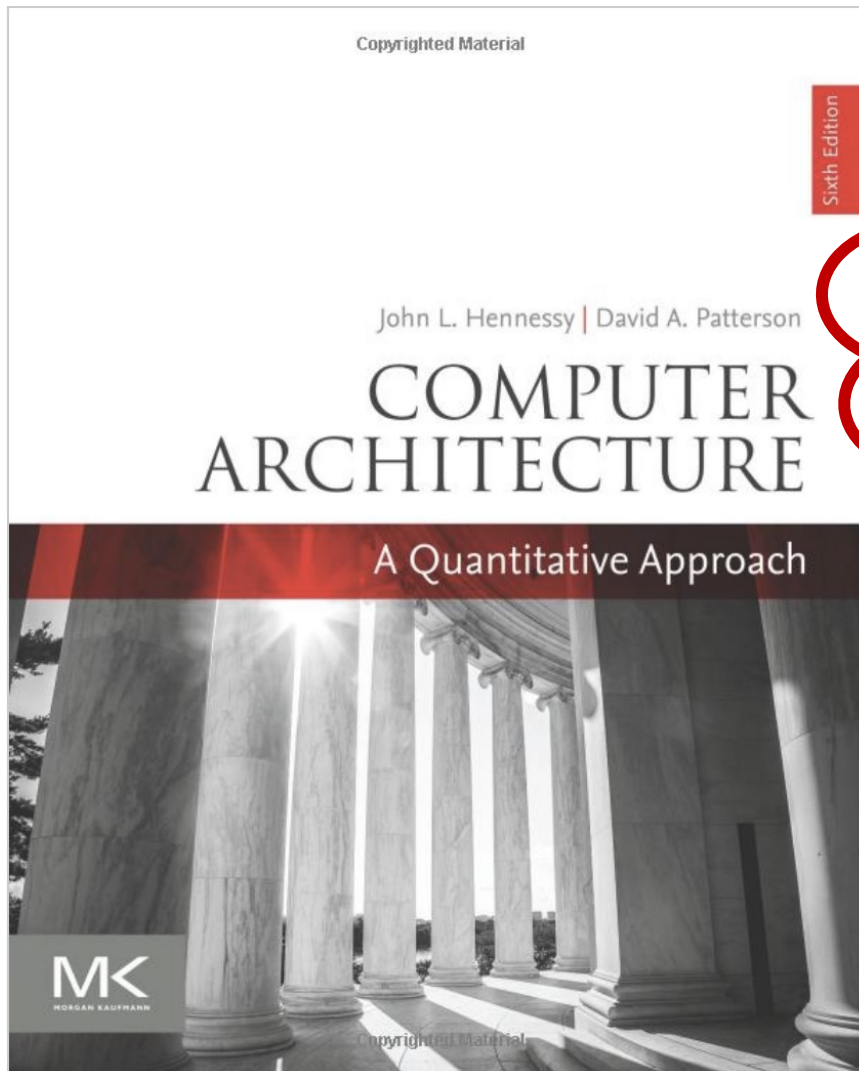


Table of Contents

Printed Text

1. Fundamentals of Quantitative Design and Analysis
2. Memory Hierarchy Design
3. Instruction-Level Parallelism and Its Exploitation
4. Data-Level Parallelism in Vector, SIMD, and GPU Architectures
5. Multiprocessors and Thread-Level Parallelism
6. The Warehouse-Scale Computer
7. Domain Specific Architectures
 - A. Instruction Set Principles
 - B. Review of Memory Hierarchy
 - C. Pipelining, Basic and Intermediate Concepts

Online

- D. Storage Systems
- E. Embedded Systems
- F. Interconnection Networks
- G. Vector Processors
- H. Hardware and Software for VLIW and EPIC
- I. Large-Scale Multiprocessors and Scientific Applications
- J. Computer Arithmetic
- K. Survey of Instruction Set Architectures
- L. Advanced Concepts on Address Translation
- M. Historical Perspectives and References

e-book...

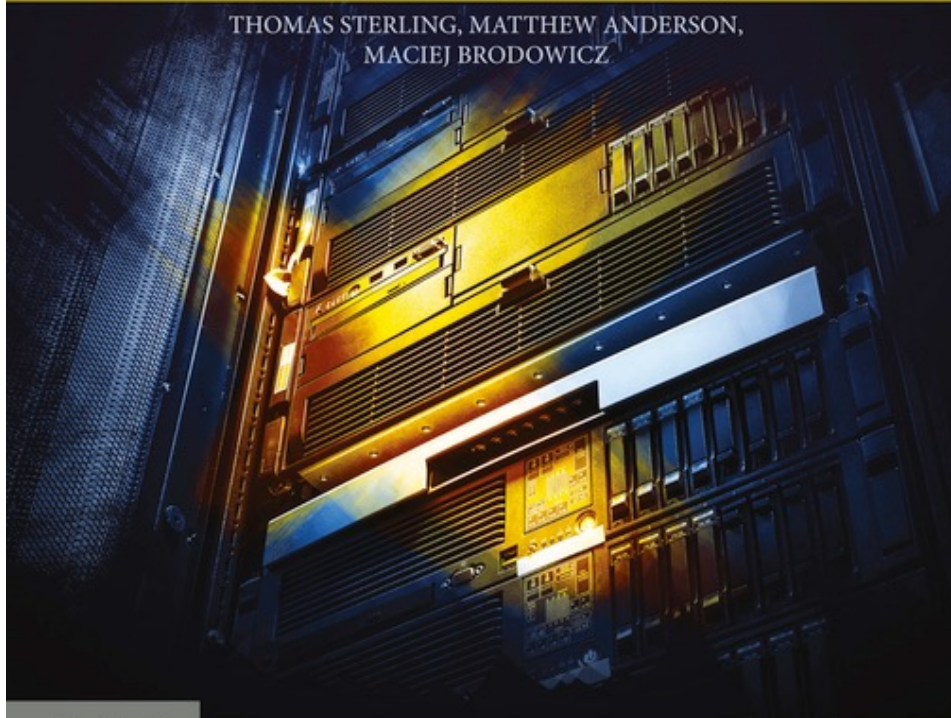
Key textbook (2)



HIGH PERFORMANCE COMPUTING

MODERN SYSTEMS AND PRACTICES

THOMAS STERLING, MATTHEW ANDERSON,
MACIEJ BRODOWICZ



MK
MORGAN KAUFMANN

FOREWORD BY C. GORDON BELL

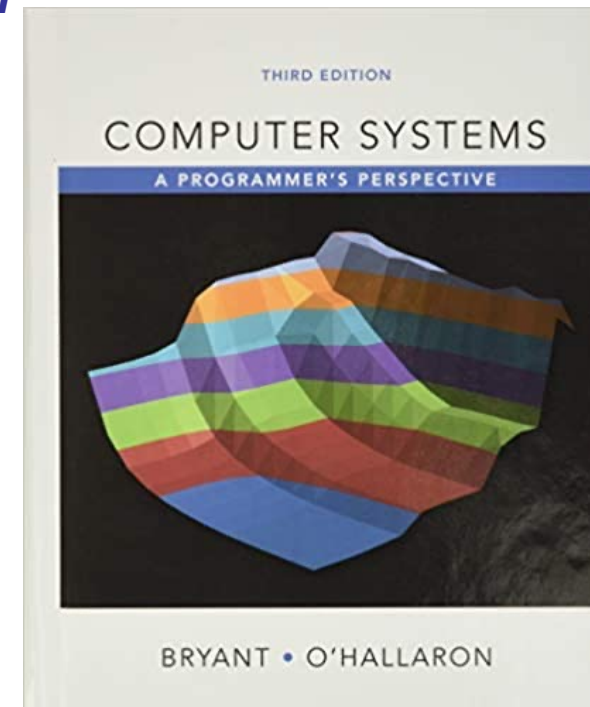
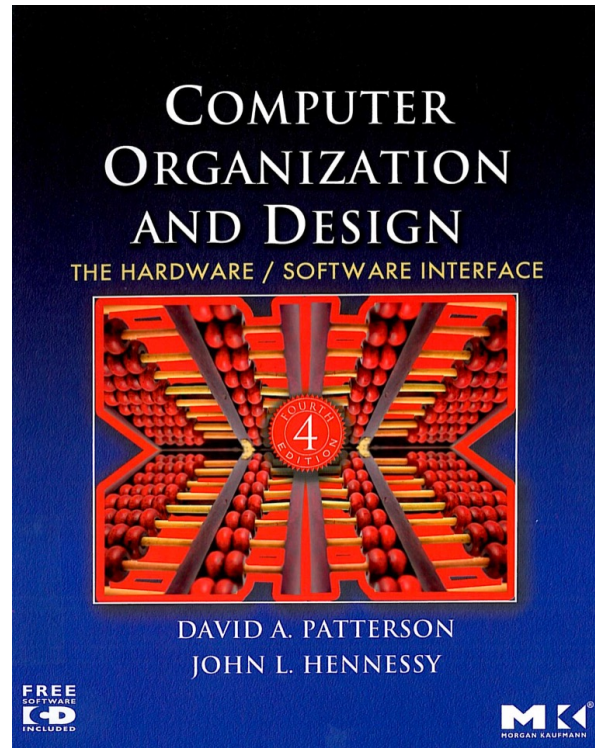
Background concepts from a basic Computer Systems course



Some notes/comments for this course:

– some slides are borrowed from

and some from



more details at
<http://gec.di.uminho.pt/miei/sc2021/>

Background for Advanced Architectures in Parallel Computing

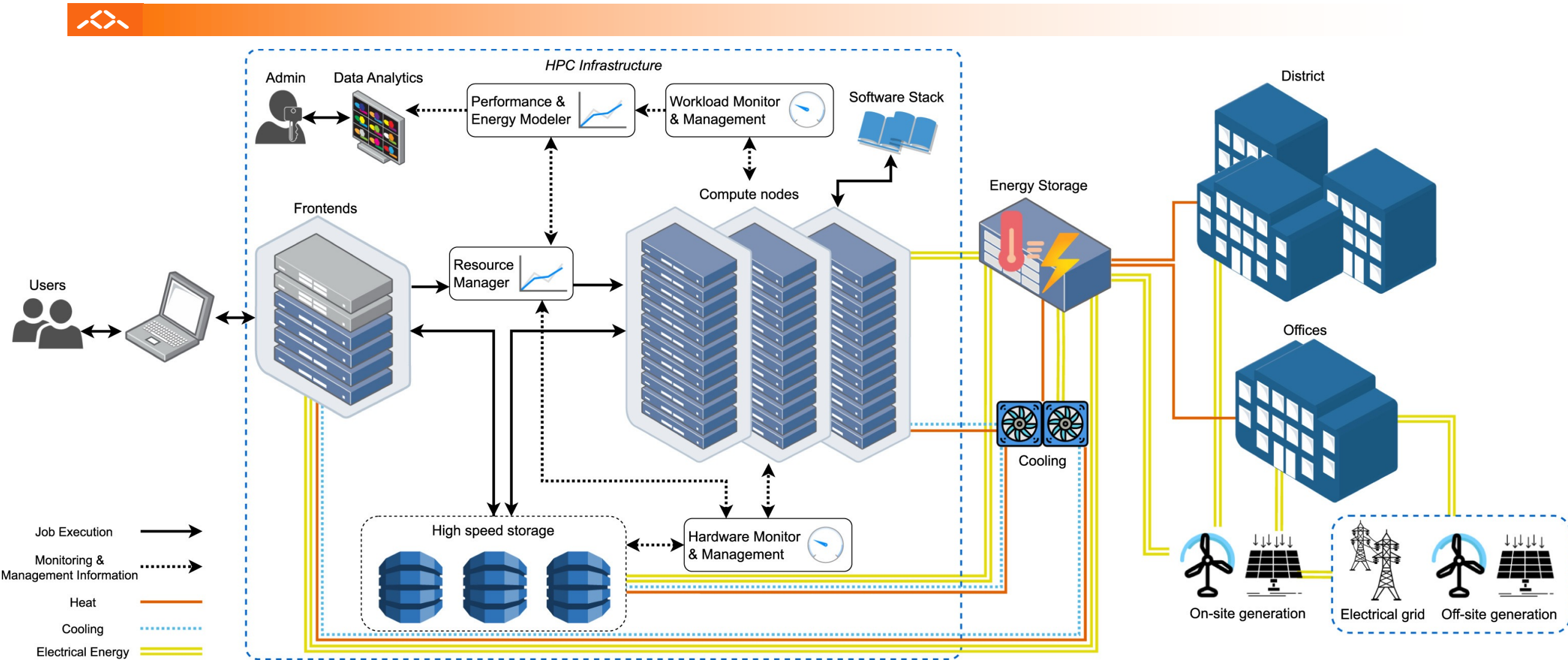


Key concepts to revise:

- *numerical data representation (integers & FP)*
- *ISA (Instruction Set Architecture)*
- *how C compilers generate code (a look into assembly code)*
 - *how scalar and structured data are allocated*
 - *how control structures are implemented*
 - *how to call/return from function/procedures*
 - *what architecture features impact performance*
- *improvements to enhance performance in a single PU*
 - *ILP: pipeline, multiple issue, ...*
 - *data parallelism: SIMD/vector processing, ...*
 - *thread-level parallelism*
 - *memory hierarchy: cache levels, ...*

Keyword: parallelism

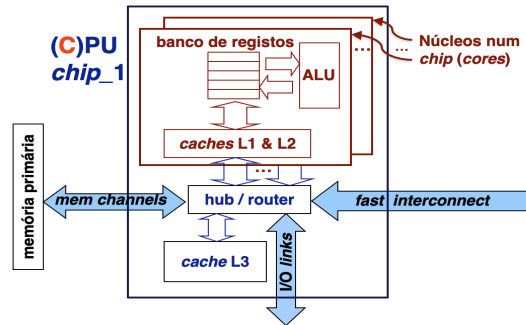
Supercomputing Architecture – an Holistic View



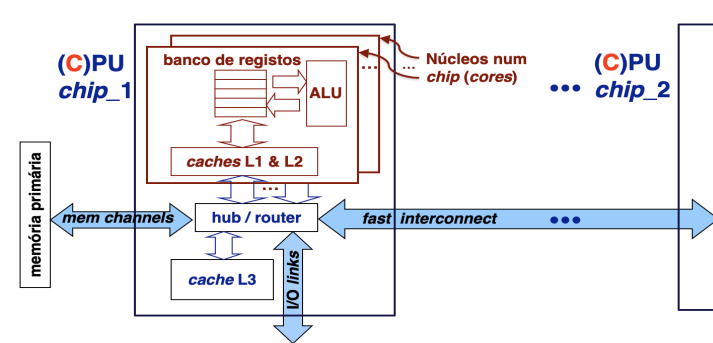
Memory systems in current computer architectures



Shared memory systems

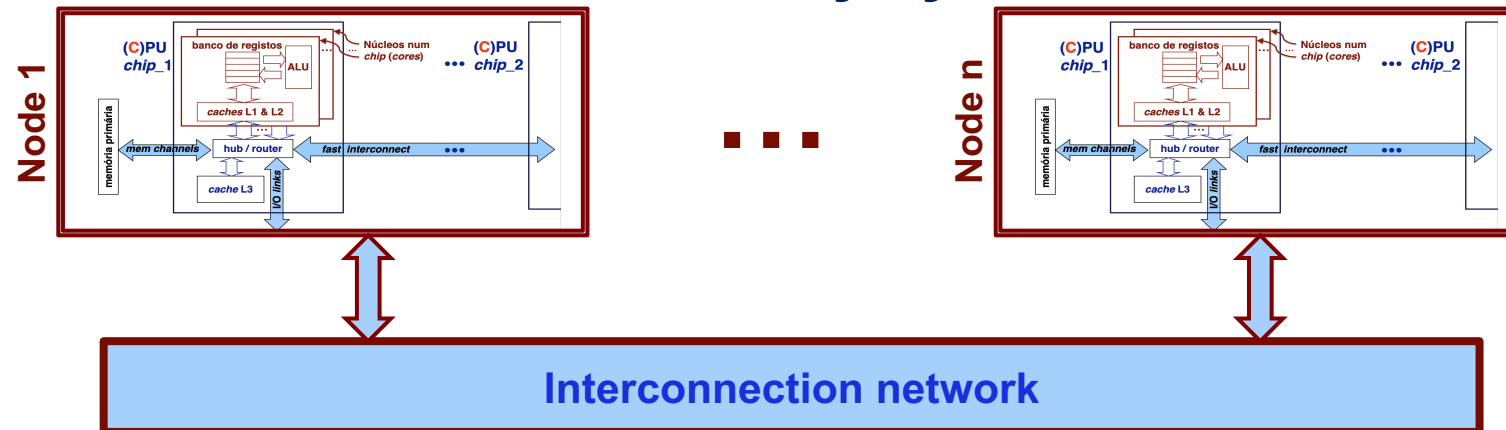


UMA: Uniform Memory Access

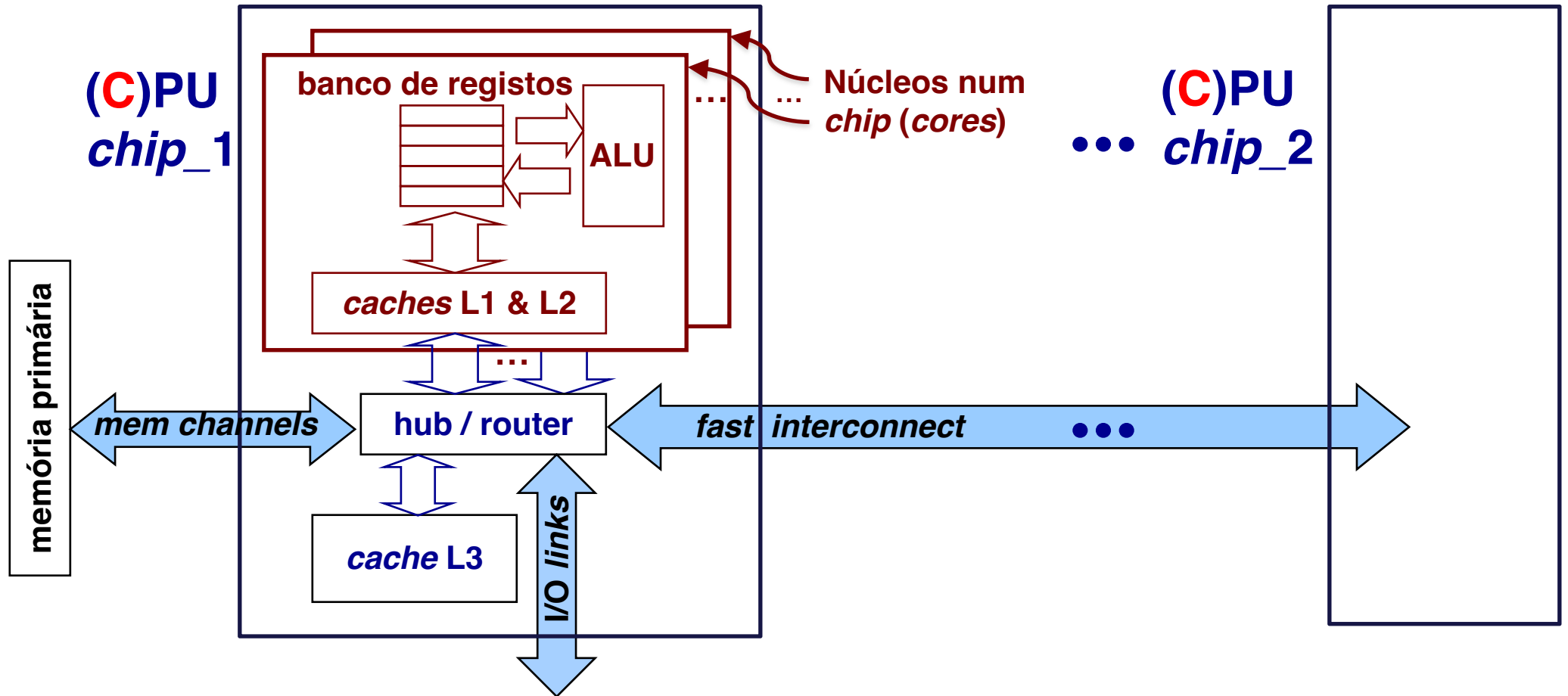


NUMA: Non-Uniform Memory Access

Distributed shared-memory systems



Multicore architecture in a server system



Revisiting....

Organização hierárquica da memória

